

Approach to Nonlinear dynamics by piecewise constant system

Jaeyoung Yoon

Seoul National University
jyoung924@snu.ac.kr

September 15, 2023

Goal

- 1 Find the **approximate solution** of **nonlinear** second order ODE.
- 2 **Continuous** approximate solution.
- 3 No guess is needed.

$$m\ddot{x} + c\dot{x} + kx = f(t, x, \dot{x}).$$

Main Idea

- ① For fixed $t > 0$, divide the interval $[0, t)$ as

$$\left[\frac{i}{N}, \frac{i+1}{N} \right), \quad i = 0, 1, \dots, [Nt],$$

where $[\cdot]$ is the floor function.

- ② On each interval, convert the ODE into the solvable form.

$$m\ddot{X}_i + c\dot{X}_i + kX_i = f\left(\frac{i}{N}, X_i\left(\frac{i}{N}\right), \dot{X}_i\left(\frac{i}{N}\right)\right), \quad t \in \left[\frac{i}{N}, \frac{i+1}{N}\right).$$

- ③ Continuity condition & Assemble

$$X(t) = X_i(t), \quad t \in \left[\frac{i}{N}, \frac{i+1}{N}\right).$$

Continuity Condition

To get the continuous approximate solution $X(t)$ and $\dot{X}(t)$, we assume the following condition:

$$d_i = X_i\left(\frac{i}{N}\right) = \lim_{s \rightarrow \frac{i}{N}} X_{i-1}(s),$$

$$v_i = \dot{X}_i\left(\frac{i}{N}\right) = \lim_{s \rightarrow \frac{i}{N}} \dot{X}_{i-1}(s).$$

Piecewise Constant Argument

Theorem

For a fixed $t > 0$,

$$\lim_{N \rightarrow \infty} \frac{[Nt]}{N} = t,$$

where $[\cdot]$ is the greatest-integer function.

Proof Since

$$[Nt] \leq Nt < [Nt] + 1,$$

we have

$$\frac{[Nt]}{N} \leq t < \frac{[Nt]}{N} + \frac{1}{N},$$

which implies the desired result. □

The simplest case

Equation:

$$\ddot{x} + \omega^2 x = 0.$$

Consider $X_i(t)$ on $\left[\frac{i}{N}, \frac{i+1}{N}\right)$, $i = 0, 1, \dots, [Nt]$, following

$$\ddot{X}_i + \omega^2 X_i \left(\frac{i}{N}\right) = 0.$$

We have

$$\dot{X}_i(t) = \int_{\frac{i}{N}}^t -\omega^2 X_i \left(\frac{i}{N}\right) ds + v_i,$$

and

$$X_i(t) = \left[1 - \frac{\omega^2}{2} \left(t - \frac{i}{N}\right)^2\right] d_i + \left(t - \frac{i}{N}\right) v_i$$

by the continuity condition.

From this, we get

$$\begin{bmatrix} d_i \\ v_i \end{bmatrix} = \begin{bmatrix} 1 - \frac{\omega^2}{2N^2} & \frac{1}{N} \\ -\frac{\omega^2}{N} & 1 \end{bmatrix} \begin{bmatrix} d_{i-1} \\ v_{i-1} \end{bmatrix}.$$

Note that

$$d_i = X \left(\frac{i}{N} \right),$$

so, for the approximated solution, it is done.

However, keep going and see what happens.

From the previous page,

$$X(t) = \begin{bmatrix} 1 - \frac{\omega^2}{2} \left(t - \frac{[Nt]}{N} \right)^2 & t - \frac{[Nt]}{N} \\ -\frac{\omega^2}{N} & 1 \end{bmatrix} \begin{bmatrix} 1 - \frac{\omega^2}{2N^2} & \frac{1}{N} \\ -\frac{\omega^2}{N} & 1 \end{bmatrix} \begin{bmatrix} d_{[Nt]} \\ v_{[Nt]} \end{bmatrix}.$$

Note that

$$\begin{bmatrix} d_{[Nt]} \\ v_{[Nt]} \end{bmatrix} = \begin{bmatrix} 1 - \frac{\omega^2}{2N^2} & \frac{1}{N} \\ -\frac{\omega^2}{N} & 1 \end{bmatrix}^{[Nt]} \begin{bmatrix} d_0 \\ v_0 \end{bmatrix}$$

and

$$\begin{bmatrix} 1 - \frac{\omega^2}{2N^2} & \frac{1}{N} \\ -\frac{\omega^2}{N} & 1 \end{bmatrix} = EDE^{-1},$$

where

$$D = \begin{bmatrix} \phi_1 & 0 \\ 0 & \phi_2 \end{bmatrix}, \quad E = \begin{bmatrix} \frac{1}{4N} \left[1 - \left(1 - \frac{16N^2}{\omega^2} \right) \right] & \frac{1}{4N} \left[1 - \left(1 - \frac{16N^2}{\omega^2} \right) \right] \\ 1 & 1 \end{bmatrix},$$

$$\phi_{1,2} = 1 - \frac{\omega^2}{4N^2} \pm \frac{\omega^2}{4N^2} \sqrt{1 - \frac{16N^2}{\omega^2}}.$$

Hence, we have

$$X_{[Nt]}(t) = \begin{bmatrix} 1 - \frac{\omega^2}{2} \left(t - \frac{[Nt]}{N} \right)^2 \\ t - \frac{[Nt]}{N} \end{bmatrix}^T E \begin{bmatrix} \phi_1^{[Nt]} & 0 \\ 0 & \phi_2^{[Nt]} \end{bmatrix} E^{-1} \begin{bmatrix} d_0 \\ v_0 \end{bmatrix}.$$

By some calculations, one can derive

$$\lim_{N \rightarrow \infty} X_{[Nt]}(t) = d_0 \cos \omega t + \frac{v_0}{\omega} \sin \omega t,$$

which is the exact solution of

$$\ddot{x} + \omega^2 x = 0.$$

Nonlinear ODE

Equation:

$$\ddot{x} + \omega^2 x = F \cos(\Omega t).$$

Consider $X_i(t)$ on $\left[\frac{i}{N}, \frac{i+1}{N}\right)$, $i = 0, 1, \dots, [Nt]$, following

$$\ddot{X}_i + \omega^2 X_i = F \cos\left(\frac{\Omega i}{N}\right),$$

with the given initial data

$$X_0(0) = d_0, \quad \dot{X}_0(0) = v_0,$$

so that we define the approximated solution

$$X^N(t) := X_i(t), \quad t \in \left[\frac{i}{N}, \frac{i+1}{N}\right).$$

Then, we obtain the explicit solution

$$X_i(t) = \left[d_i - \frac{F}{\omega^2} \cos \left(\frac{\Omega i}{N} \right) \right] \cos \left(\omega \left(t - \frac{i}{N} \right) \right) \\ + v_i \sin \left(\omega \left(t - \frac{i}{N} \right) \right) + \frac{F}{\omega^2} \cos \left(\frac{\Omega i}{N} \right),$$

which induces

$$\begin{bmatrix} d_{i+1} \\ v_{i+1} \end{bmatrix} = T \begin{bmatrix} d_i \\ v_i \end{bmatrix} + \frac{F}{\omega^2} \cos \left(\frac{\Omega i}{N} \right) R,$$

where

$$T := \begin{bmatrix} \cos \frac{\omega}{N} & \frac{1}{\omega} \sin \frac{\omega}{N} \\ -\omega \sin \frac{\omega}{N} & \cos \frac{\omega}{N} \end{bmatrix}, \quad R := \begin{bmatrix} 1 - \cos \frac{\omega}{N} \\ \omega \sin \frac{\omega}{N} \end{bmatrix}.$$

Let's keep going. This time, we get

$$\begin{bmatrix} d_{[Nt]} \\ v_{[Nt]} \end{bmatrix} = T^{[Nt]} \begin{bmatrix} d_0 \\ v_0 \end{bmatrix} + \sum_{j=1}^{[Nt]} \frac{F}{\omega^2} \cos \left(\frac{\Omega([Nt] - j)}{N} \right) T^{j-1} R,$$

hence,

$$\begin{bmatrix} X_{[Nt]}(t) \\ \dot{X}_{[Nt]}(t) \end{bmatrix} = \begin{bmatrix} \cos \left(\omega \left(t - \frac{[Nt]}{N} \right) \right) \\ \sin \left(\omega \left(t - \frac{[Nt]}{N} \right) \right) \end{bmatrix}^T \cdot \begin{bmatrix} d_{[Nt]} \\ v_{[Nt]} \end{bmatrix} + \frac{F}{\omega^2} \cos \left(\frac{\Omega[Nt]}{N} \right) \left(1 - \cos \left[\omega \left(t - \frac{[Nt]}{N} \right) \right] \right).$$

Now letting N to infinity,

$$\lim_{N \rightarrow \infty} X_{[Nt]}(t) = d_0 \cos \omega t + \frac{v_0}{\omega} \sin \omega t + \frac{F}{\omega^2 - \Omega^2} (\cos \Omega t - \cos \omega t),$$

which is the exact solution to

$$\ddot{x} + \omega^2 x = F \cos(\Omega t).$$

Example

Equation:

$$\begin{cases} \ddot{x} + 4x = \cos 3t, & t > 0, \\ x(0) = 0, & \dot{x}(0) = 1. \end{cases}$$

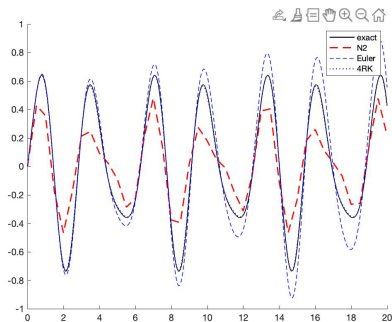
Consider for each interval $\left[\frac{i}{N}, \frac{i+1}{N}\right)$, $i = 0, 1, \dots, [Nt]$,

$$\ddot{X}_i + 4X_i = \cos \frac{3i}{N},$$

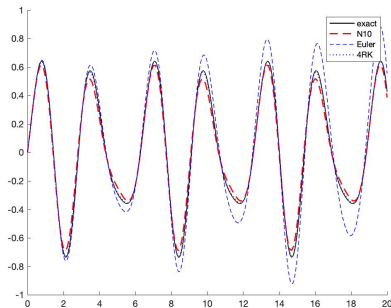
with the continuity condition. Then, we can obtain

$$\begin{bmatrix} d_{i+1} \\ v_{i+1} \end{bmatrix} = \begin{bmatrix} \cos \frac{2}{N} & \frac{1}{2} \sin \frac{2}{N} \\ -2 \sin \frac{2}{N} & \cos \frac{2}{N} \end{bmatrix} \begin{bmatrix} d_i \\ v_i \end{bmatrix} + \frac{3}{4} \cos \frac{3i}{N} \begin{bmatrix} 1 - \cos \frac{2}{N} \\ 2 \end{bmatrix}.$$

Example

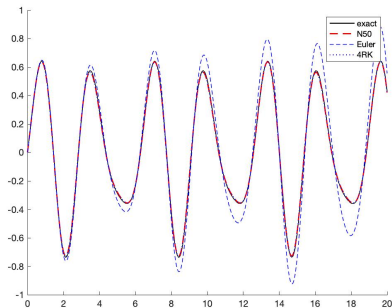


(a) PC with $N = 2$

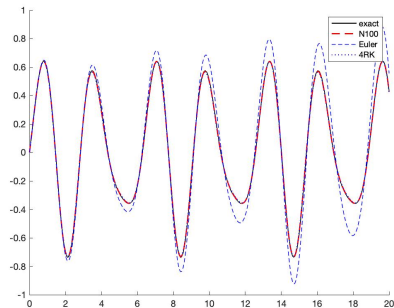


(b) PC with $N = 10$

Example



(c) PC with $N = 50$



(d) PC with $N = 100$

Example

Euler	0.2258
4th Runge-Kutta	0.0011
N=2	0.6696
N=10	0.0282
N=50	0.0011
N=100	0.0002

Table: $\|x_{\text{ex}} - X\|_{l^2([0,20])}$

Piecewise Taylor Expansion Method(PT)

For a nonlinear function f , consider

$$\ddot{x} + c\dot{x} + kx = f(t, x, \dot{x}).$$

Then,

$$\ddot{x}(t) = f(t, x(t), \dot{x}(t)) - c\dot{x}(t) - kx(t) =: g(t).$$

On the interval $\left[\frac{i}{N}, \frac{i+1}{N}\right)$, ($i = 0, 1, \dots, [Nt]$), for $0 \leq h < \frac{1}{N}$,

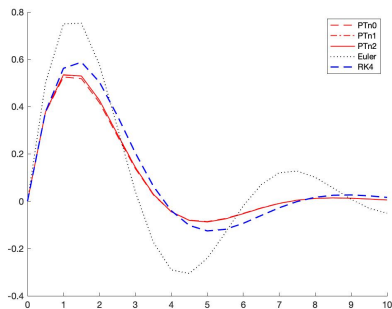
$$\ddot{X}_i\left(\frac{i}{N} + h\right) = g\left(\frac{i}{N} + h\right) \approx \sum_{j=0}^n \frac{h^j}{j!} g^{(j)}\left(\frac{i}{N}\right)$$

With the continuity condition, we can easily get

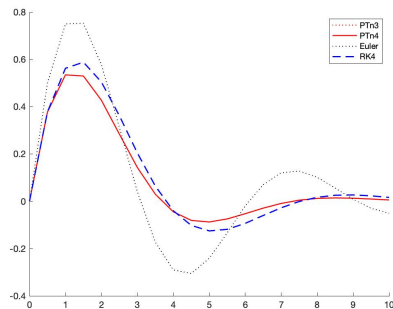
$$X_i\left(\frac{i}{N} + h\right) = d_i + v_i h + \sum_{j=0}^n \frac{h^{j+2}}{(j+2)!} g^{(j)}\left(\frac{i}{N}\right).$$

Example1

$$\ddot{x} + \dot{x} + x = \frac{1}{10}x^3$$



(e) PT with $n = 0, 1, 2$

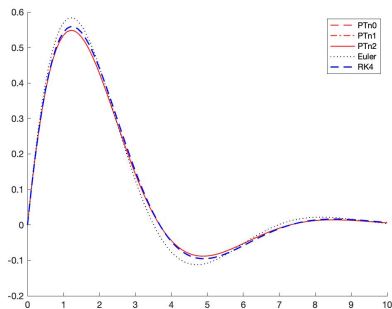


(f) PT with $n = 3, 4$

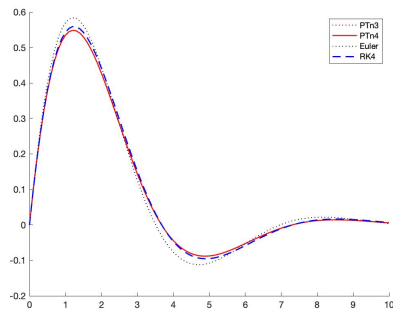
Figure: $N = 2$ ($\Delta t = 0.5$)

Example1

$$\ddot{x} + \dot{x} + x = \frac{1}{10}x^3$$



(a) PT with $n = 0, 1, 2$



(b) PT with $n = 3, 4$

Figure: $N = 10$ ($\Delta t = 0.1$)

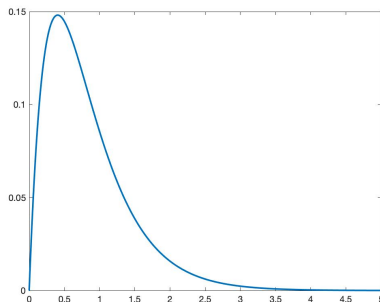
Comparison with Other Methods

Consider

$$\ddot{x} + 5\dot{x} + 6x = 0.$$

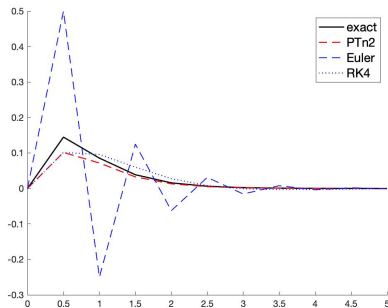
We have the explicit solution,

$$x_{\text{ex}}(t) = (3d_0 + v_0) e^{-2t} - (2d_0 + v_0) e^{-3t}.$$

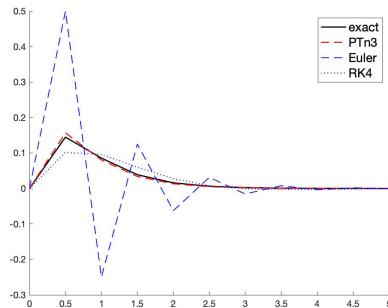


Example2(i)

$$N = 2 \ (\Delta t = 0.5)$$



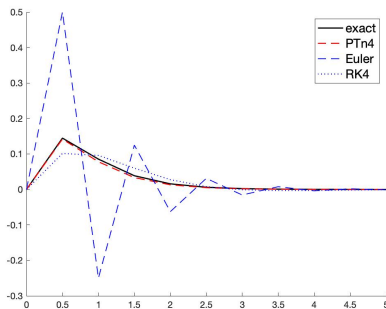
(a) PT with $n = 2$



(b) PT with $n = 3$

Example2(i)

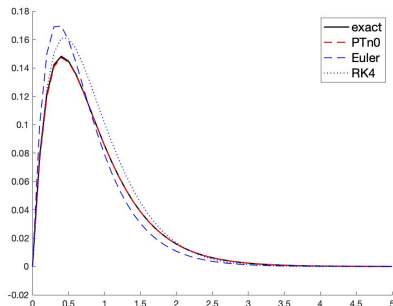
$$N = 2 \ (\Delta t = 0.5)$$



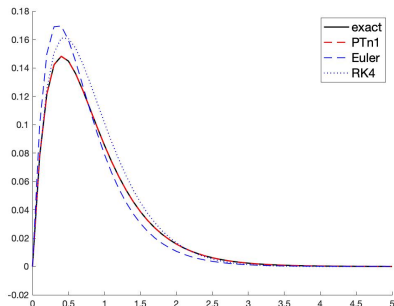
(c) PT with $n = 4$

Example2(ii)

$$N = 10 \ (\Delta t = 0.1)$$



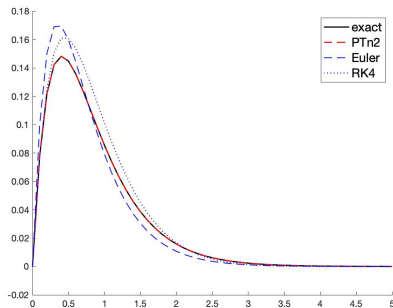
(d) PT with $n = 0$



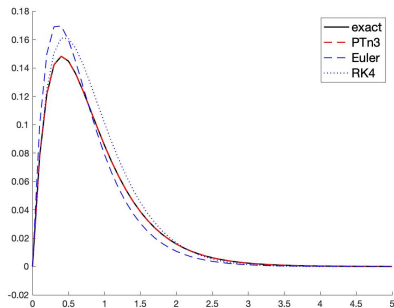
(e) PT with $n = 1$

Example2(ii)

$$N = 10 \ (\Delta t = 0.1)$$



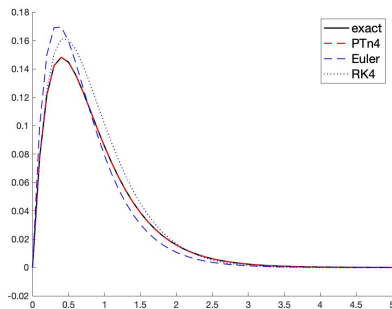
(f) PT with $n = 2$



(g) PT with $n = 3$

Example2(ii)

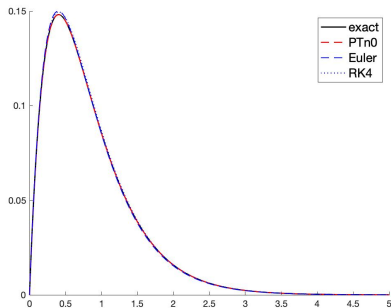
$$N = 10 \ (\Delta t = 0.1)$$



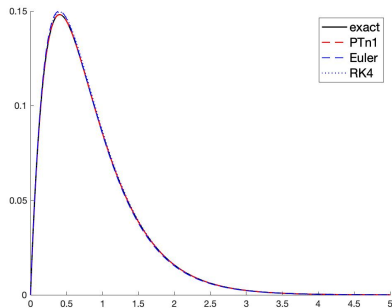
(h) PT with $n = 4$

Example2(iii)

$$N = 100 \ (\Delta t = 0.01)$$



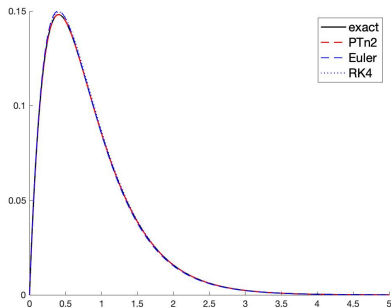
(i) PT with $n = 0$



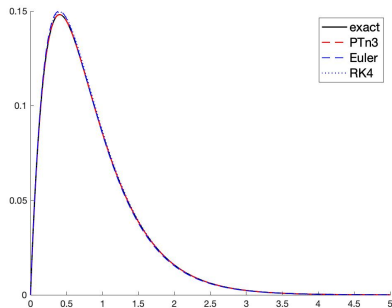
(j) PT with $n = 1$

Example2(iii)

$$N = 100 \ (\Delta t = 0.01)$$



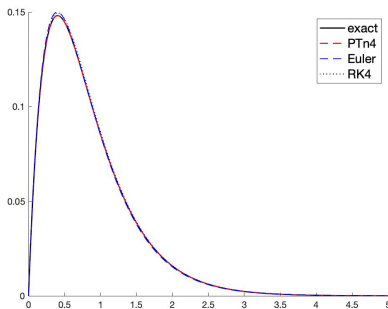
(k) PT with $n = 2$



(l) PT with $n = 3$

Example2(iii)

$$N = 100 \ (\Delta t = 0.01)$$



(m) PT with $n = 4$

Example2 Result

	Euler	4th-RK	PTn0	PTn1
N=2	0.1267	0.0013	-	-
N=10	3.2e-4	2.6e-4	9.4e-4	2.2e-5
N=100	2.5e-6	2.2e-6	9.4e-6	1.4e-9

	PTn2	PTn3	PTn4
N=2	0.0042	0.0011	4.9e-5
N=10	1.4e-7	6.0e-10	1.6e-12
N=100	1.0e-13	4.3e-18	1.1e-22

Table: $\|x_{\text{ex}} - X\|_{l^2([0,5])}$

Note that

$$\Delta t = \frac{1}{N}.$$

Remark

- ① Piecewise constant system exhibits good accuracy compared to 4th Runge Kutta scheme.
- ② With the low order of Taylor expansion, we can have good accuracy with a big time step.

Can we apply PC/PT method to dynamical system?

Example Cucker-Smale model

$$\begin{aligned}\dot{x}_i &= v_i, \quad i = 1, \dots, M, \\ \dot{v}_i &= \frac{\kappa}{M} \sum_{j=1}^M \phi(\|x_j - x_i\|) (v_j - v_i).\end{aligned}$$

For the simplest case, $\phi \equiv 1$, on interval $\left[\frac{r}{N}, \frac{r+1}{N}\right)$, $r = 0, 1, \dots, [Nt]$,

$$\dot{v}_i^r(t) = \frac{\kappa}{M} \sum_{j=1}^M (\psi_j^r - \psi_i^r),$$

where

$$\psi_j^r = \lim_{s \rightarrow \frac{r}{N}} v_j^{r-1}(s) = v_i^r\left(\frac{r}{N}\right).$$

We have

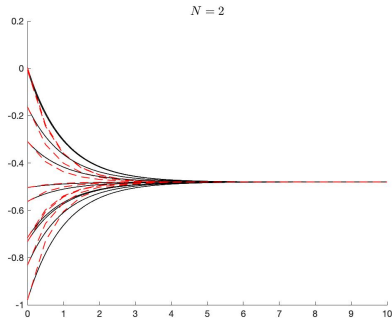
$$v_i^r(t) = \left(t - \frac{r}{N}\right) \frac{\kappa}{M} \sum_{j=1}^M (\psi_j^r - \psi_i^r) + \psi_i^r,$$

hence,

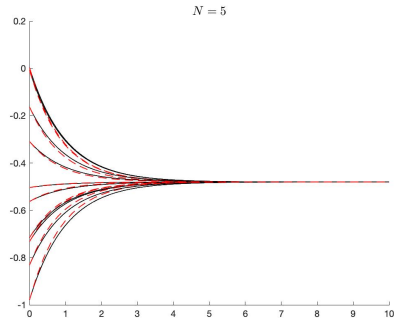
$$\begin{bmatrix} \psi_1^r \\ \vdots \\ \psi_M^r \end{bmatrix} = \begin{bmatrix} 1 - \frac{\kappa(N-1)}{MN} & * & \cdots & * \\ * & 1 - \frac{\kappa(N-1)}{MN} & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ * & \cdots & * & 1 - \frac{\kappa(N-1)}{MN} \end{bmatrix} \begin{bmatrix} \psi_1^{r-1} \\ \vdots \\ \psi_M^{r-1} \end{bmatrix},$$

where

$$* = \frac{\kappa}{NM}.$$



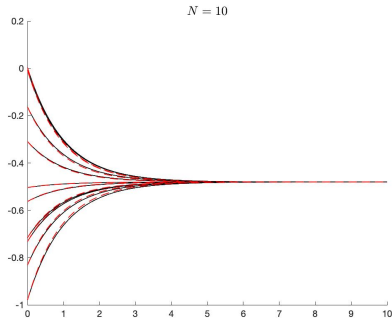
(n) PC with $N = 2$ (red)



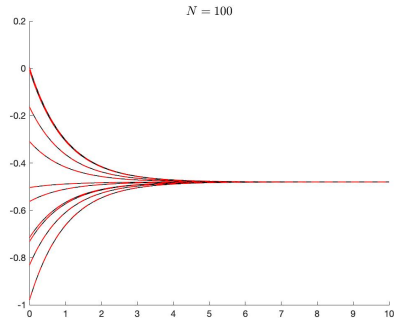
(o) PC with $N = 5$ (red)

4RK	PCN2	PCN5	PCN10	PCN100
0.0054	0.0006	0.0010	0.0007	0.0018

Table: Computation time (Sec)



(p) PC with $N = 10$ (red)



(q) PC with $N = 100$ (red)

4RK	PCN2	PCN5	PCN10	PCN100
0.0054	0.0006	0.0010	0.0007	0.0018

Table: Computation time (Sec)

We can also apply to Kuramoto model, Winfree model, etc. (PT method)

Kuramoto model

$$\dot{\theta}_i = \frac{\kappa}{M} \sum_{j=1}^M \sin(\theta_j - \theta_i).$$

Winfree model

$$\dot{\theta}_i = \frac{\kappa}{M} \sum_{j=1}^M S(\theta_i) I(\theta_j).$$

Thank you for listening!